BEYOND CHATGPT

How Intelligent Software Systems Think and Work

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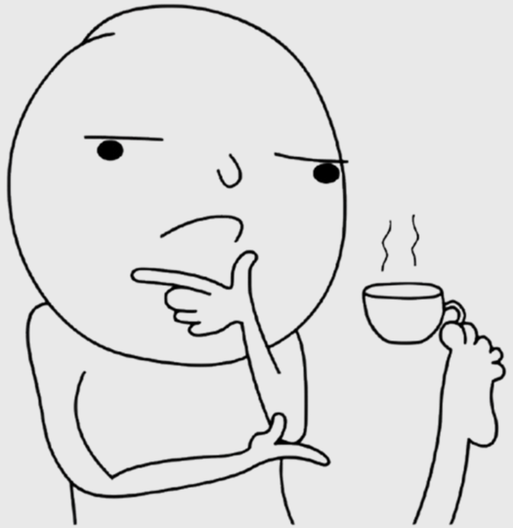
WHAT IS CHATGPT ACTUALLY?

When you type a question into ChatGPT, you are talking to:

- A magical AI brain
- A super-intelligent robot
- A software system made of multiple parts
- Something only geniuses can understand



WHY DID THIS AI MESS UP?

1. **User:** “What’s your refund policy for damaged items?”

   AI: “Returns are accepted within 30 days for unused items in original packaging.”

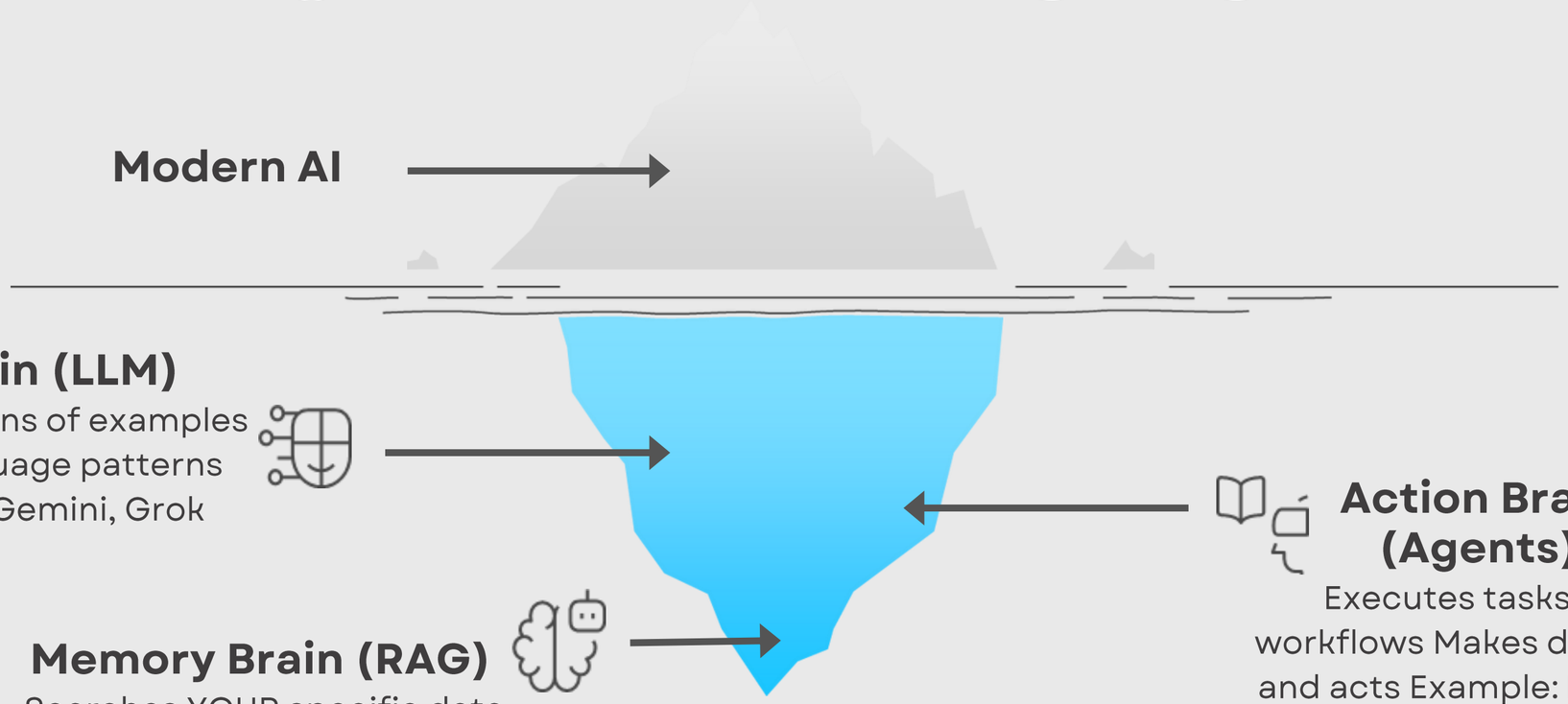
   User: “That doesn’t answer my question.”

2. **User:** “Do you ship to Bangalore?”

   AI: “Yes! We ship worldwide to over 100 countries across 6 continents…”

   User: “Bro. I asked YES or NO.”

THE THREE BRAINS OF MODERN AI SYSTEMS



Pattern Brain (LLM)
Learned from millions of examples
Understands language patterns
Example: GPT-5, Gemini, Grok

Memory Brain (RAG)
Searches YOUR specific data
Retrieves relevant information
Example: Company knowledge bases

Action Brain (Agents)
Executes tasks and workflows Makes decisions and acts Example: Booking systems, automation

THE REAL SHIFT

2020–2021:
"Wait... software can talk?" 🤯

2022–2023:
"Add ChatGPT everywhere!"

2024:
Reality check.

2025–2026:
The Actual Shift,
Companies stopped asking: **"How do we use ChatGPT?"**
Started asking: **"How do we build AI systems that work for us?"**

The Problem:
ChatGPT alone doesn't know YOUR data or workflows

The Solution:
LLMs + RAG + Agents = Custom AI systems

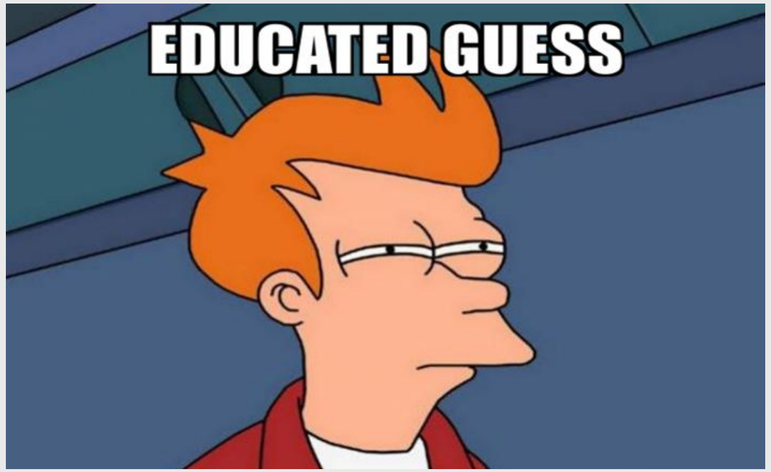
Market Reality:

- **ChatGPT user:** Common
- **RAG builder:** IN DEMAND
- **Agent builder:** RARE & HIGHLY PAID

HOW LLMS GENERATE TEXT

LLMs don't:

- Plan the whole response
- Know the ending



LLMs do:

- Generate word by word
- Each word based on previous words
- Predict → Add → Repeat
- Billions of predictions per second.

This is why:

- Responses drift off-topic
- Confident but wrong
- "Sounds right" ≠ "is right"

THE STEPS BEHIND EVERY AI RESPONSE

Step 1:
Tokenization of Input

Step 2:
Embeddings creation

Step 3:
Neural network Processing

Step 4:
Probability calculation

Step 5:
Repeat (Continue until complete)

This happens in milliseconds.
Pure math. Pattern recognition at massive scale.

THE FUNDAMENTAL PROBLEM

LLMs are great at language but, CANNOT "**Access YOUR documents**" or **"Know recent information"**

Why?

They predict what **SOUNDS right**, not what **IS right**.

How do we connect LLMs to YOUR actual data?

RAG

(RETRIEVAL-AUGMENTED GENERATION)

Before RAG:

User: "What's our leave policy?"

LLM: "Most companies offer 15-20 days..." ❌

After RAG:

User: "What's our leave policy?"

System: [Searches]

LLM: "You get 24 paid leaves annually" ✅

Search YOUR documents first, then let LLM answer.

BUT HOW DO WE SEARCH BY MEANING?

Old Way (Keyword Matching)

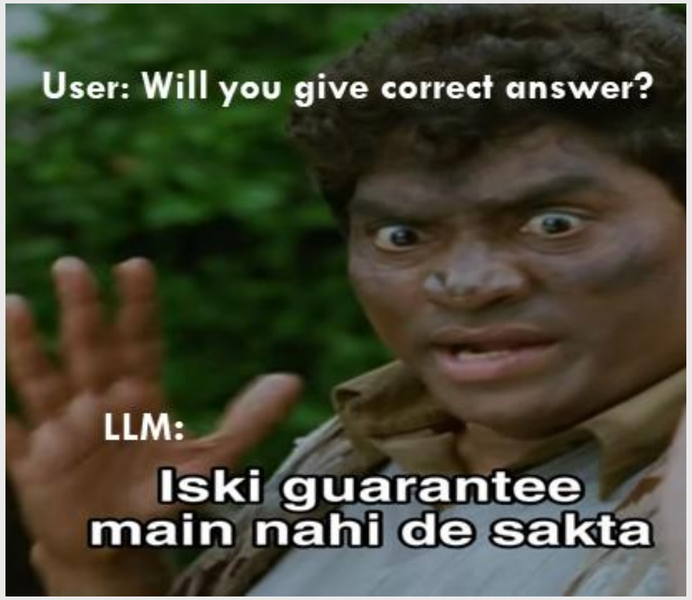
- Searches exact words

Rigid, often incomplete

Smart Way(Semantic Search)

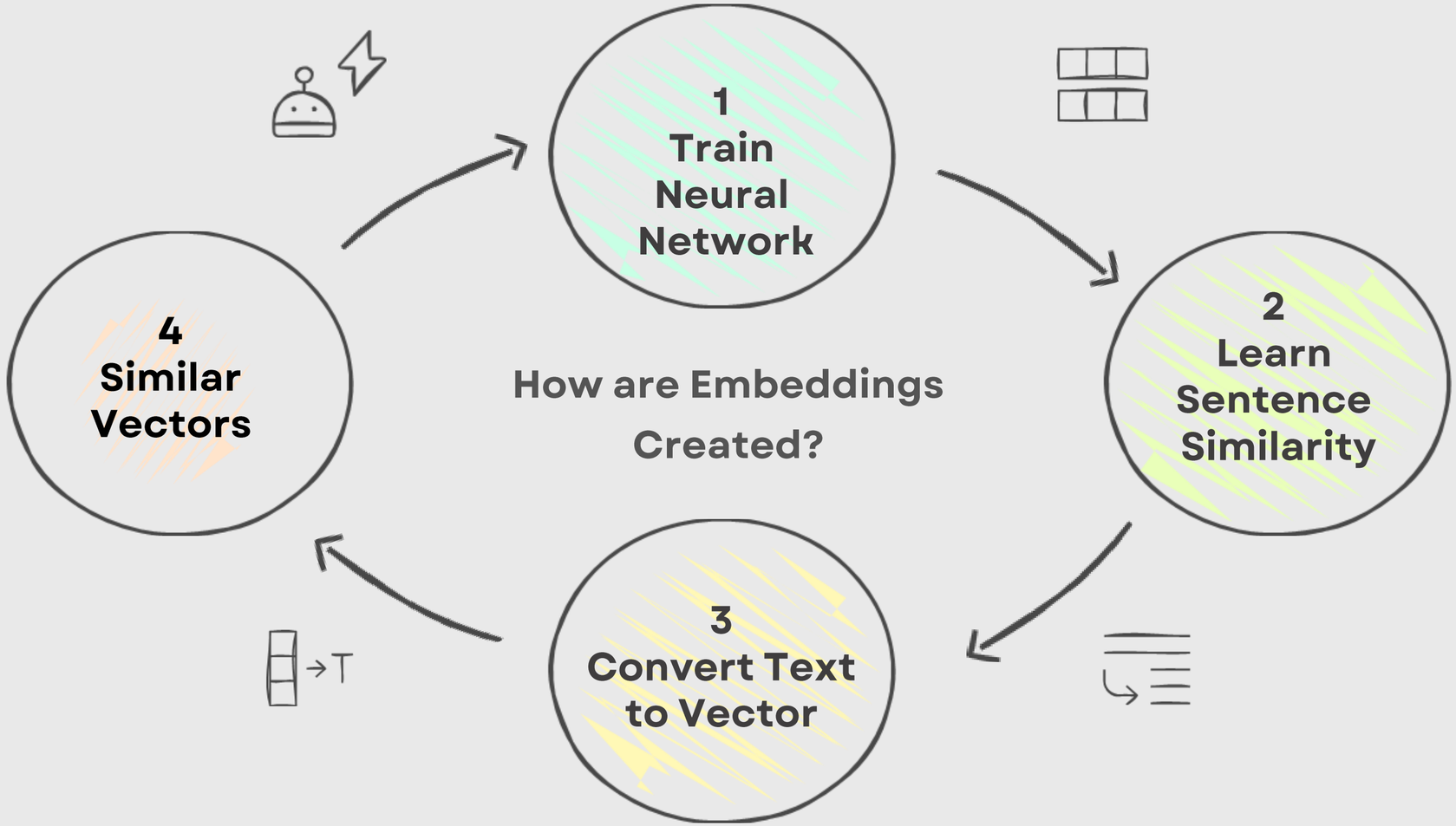
- Understands meaning, not words

Finds what you actually want



How does a computer understand "meaning"?

VECTORS & EMBEDDINGS



AGENTS

(THE ACTION TAKERS)

User:

"Book me the cheapest flight to Goa next Friday under ₹5000“

LLM alone:

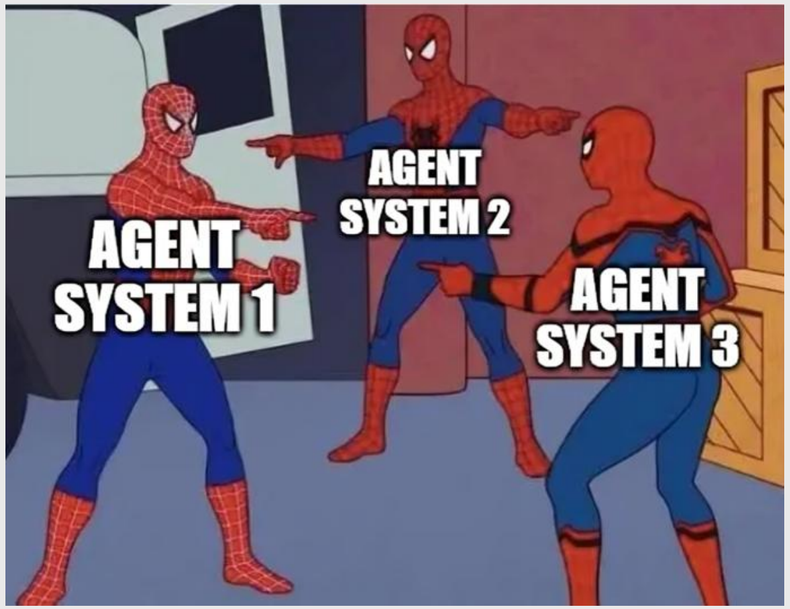
"Here's how you can search for flights..."

(just advice)

Agent:

- Searches flight APIs
- Compares prices
- Checks your calendar for conflicts
- Books the ticket
- Sends confirmation

MULTI-AGENT SYSTEMS



The principle:
Specialized agents
is better than
One generalist agent

USER: "Grow my Instagram to 100k followers"

MULTI-AGENT ORCHESTRATION:

Agent 1: COORDINATOR AGENT
Plans & orchestrates

Agent 2: TRENDS AGENT
Monitors viral trends, Analyzes competitors, Suggests topics

Agent 3: CONTENT CREATOR
Creates reels, edits videos, Designs Posts

Agent 4: CAPTION WRITER
Writes hooks, Adds hashtags

GUARDRAILS & SAFETY

POWERFUL TOOLS NEED SAFETY NETS

Small errors
=
Big damage.

The Balance

Too strict: Useless

Too loose: Dangerous

Good agents = Powerful + Safe

Essential Guardrails:

Hard Limits: Max API calls, cost caps, time limits

Permissions: Read-only access, human approval for critical actions

Monitoring: Track actions, alert on unusual patterns, rollback ability

CAREER OPPORTUNITIES IN MODERN AI

Salary (India, rough ranges)

- Entry-level AI / Backend: **₹5–8 LPA**
- Mid-level (2–4 yrs): **₹12–25 LPA**
- Strong system engineers: **₹30L+**

(varies by company & skill, not just degree)

Computer Science students, After getting a Job in AI

THANK YOU

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